

PATTERN PRINTING

Write a Java program to print a right triangle star pattern.

```
public class Pattern 1 {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print("* ");  
                System.out.println();  
            }  
        }  
    }  
}
```

Explanation:

Outer Loop (Rows)

Code: for (int i = 1; i <= 5; i++)

Purpose: Controls the total number of rows.

Execution: It runs exactly 5 times, creating 5 horizontal rows.

Variable i: Represents the current row number (from 1 to 5).

Inner Loop (Stars per Row)

Code: for (int j = 1; j <= i; j++)

Purpose: Controls how many stars (*) are printed in each row.

Execution: It depends entirely on the current value of the row variable i.

Action:

Row 1 (i = 1): Inner loop runs 1 time \(\rightarrow\) Prints 1 star.

Row 2 (i = 2): Inner loop runs 2 times \(\rightarrow\) Prints 2 stars.

Row 5 (i = 5): Inner loop runs 5 times \(\rightarrow\) Prints 5 stars.

Printing and Newlines

System.out.print("* ");

Prints a star followed by a space.

Keeps the cursor on the same line so stars print side-by-side.

System.out.println();

Moves the cursor to the next line.

Runs after the inner loop finishes printing all stars for the current row.

Final Output

```
*  
  
* *  
  
* * *  
  
* * * *  
  
* * * * *
```

Nester for loop

```
public class MultiplicationTable {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 3; i++) {  
            for (int j = 1; j <= 3; j++) {  
                System.out.print(i * j + "\t");  
                System.out.println();  
            }  
        }  
    }  
}
```

Explanation

Outer Loop: Controls the rows using variable i from 1 to 3.

Inner Loop: Controls the columns using variable j from 1 to 3.

Calculation: Multiplies the current row i by column j.

Spacing: Uses \t to insert a tab space between numbers.

Line Break: Uses println() to start a new row after the inner loop finishes.

Execution Step-by-Step

Row 1 (i = 1):

Multiplies 1 by 1, 2, and 3.

Prints: 1 2 3

Row 2 (i = 2):

Multiplies 2 by 1, 2, and 3.

Prints: 2 4 6

Row 3 (i = 3):

Multiplies 3 by 1, 2, and 3.

Prints: 3 6 9

Output

```
1      2      3
2      4      6
3      6      9
```

Pattern using for loop

Write a Java program to print an inverted right triangle star pattern.

```
public class Pattern2 {
    public static void main(String[] args) {
        for (int i = 5; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("** ");
            }
            System.out.println();
        }
    }
}
```

Code Explanation

for (int i = 5; i >= 1; i--) (Outer Loop): This loop controls the total number of rows. It starts at 5 and decreases by 1 after each row until it reaches 1. This means the program will print exactly 5 rows.

for (int j = 1; j <= i; j++) (Inner Loop): This loop controls how many stars (*) are printed in each row. The number of stars printed is determined by the current value of i.

Output

* * * * *

* * * *

* * *

* *

*